

Foundations of RoBERTa

[RoBERTa]

1.0 Content Block

History BERT was originally published by Google researchers Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. The design has its origins from pre-training contextual representations, including semi-supervised sequence learning, generative pre-training, ELMo, and ULMFit. Unlike previous models, BERT is a deeply bidirectional, unsupervised language representation, pre-trained using only a plain text corpus. Context-free models such as word2vec or GloVe generate a single word embedding representation for each word in the vocabulary, whereas BERT takes into account the context for each occurrence of a given word. For instance, whereas the vector for "running" will have the same word2vec vector representation for both of its occurrences in the sentences "He is running a company" and "He is running a marathon", BERT will provide a contextualized embedding that will be different according to the sentence. On October 25, 2019, Google announced that they had started applying BERT models to English-language search queries on Google Search within the US. On December 9, 2019, it was reported that BERT had been adopted by Google Search for over 70 languages. In October 2020, almost every single English-based query was processed by a BERT model.

2.0 Content Block

Embedding This section describes the embedding used by BERTBASE. The other one, BERTLARGE, is similar, just larger. The tokenizer of BERT is WordPiece, which is a sub-word strategy like byte-pair encoding. Its vocabulary size is 30,000, and any token not appearing in its vocabulary is replaced by [UNK] ("unknown").

3.0 Content Block

In masked language modeling, 15% of tokens would be randomly selected for masked-prediction task, and the training objective was to predict the masked token given its context. In more detail, the selected token is:

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Cost BERT was trained on the BookCorpus (800M words) and a filtered version of English Wikipedia (2,500M words) without lists, tables, and headers. Training BERTBASE on 4 cloud TPU (16 TPU chips total) took 4 days, at an estimated cost of 500 USD. Training BERTLARGE on 16 cloud TPU (64 TPU chips total) took 4 days.

5.0 Content Block

Bidirectional encoder representations from transformers (BERT) is a language model introduced in October 2018 by researchers at Google. It learns to represent text as a sequence of vectors using self-supervised learning. It uses the encoder-only transformer architecture. BERT dramatically improved the state of the art for large language models. As of 2020, BERT is a ubiquitous baseline in natural language processing (NLP) experiments. BERT is trained by masked token prediction and next sentence prediction. With this training, BERT learns contextual, latent representations of tokens in their context, similar to ELMo and GPT-2. It found applications for many natural language processing tasks, such as coreference resolution and polysemy resolution. It improved on ELMo and spawned the study of "BERTology", which attempts to interpret what is learned by BERT. BERT was originally implemented in the English language at two model sizes, BERTBASE (110 million parameters) and BERTLARGE (340 million parameters). Both were trained on the Toronto BookCorpus (800M words) and English Wikipedia

(2,500M words). The weights were released on GitHub. On March 11, 2020, 24 smaller models were released, the smallest being BERTTINY with just 4 million parameters.

6.0 Content Block

BERT is meant as a general pretrained model for various applications in natural language processing. That is, after pre-training, BERT can be fine-tuned with fewer resources on smaller datasets to optimize its performance on specific tasks such as natural language inference and text classification, and sequence-to-sequence-based language generation tasks such as question answering and conversational response generation. The original BERT paper published results demonstrating that a small amount of finetuning (for BERTLARGE, 1 hour on 1 Cloud TPU) allowed it to achieved state-of-the-art performance on a number of natural language understanding tasks:

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Given "[CLS] my dog is cute [SEP] he likes playing [SEP]", the model should predict [IsNext]. Given "[CLS] my dog is cute [SEP] how do magnets work [SEP]", the model should predict [NotNext].

8.0 Content Block

the feed-forward size and filter size are synonymous. Both of them denote the number of dimensions in the middle layer of the feed-forward network. the hidden size and embedding size are synonymous. Both of them denote the number of real numbers used to represent a token. The notation for encoder stack is written as L/H. For example, BERTBASE is written as 12L/768H, BERTLARGE as 24L/1024H, and BERTTINY as 2L/128H.

9.0 Content Block

Variants The BERT models were influential and inspired many variants. RoBERTa (2019) was an engineering improvement. It preserves BERT's architecture (slightly larger, at 355M parameters), but improves its training, changing key hyperparameters, removing the next-sentence prediction task, and using much larger mini-batch sizes. XLM-RoBERTa (2019) was a multilingual RoBERTa model. It was one of the first works on multilingual language modeling at scale. DistilBERT (2019) distills BERTBASE to a model with just 60% of its parameters (66M), while preserving 95% of its benchmark scores. Similarly, TinyBERT (2019) is a distilled model with just 28% of its parameters. ALBERT (2019) used shared-parameter across layers, and experimented with independently varying the hidden size and the word-embedding layer's output size as two hyperparameters. They also replaced the next sentence prediction task with the sentence-order prediction (SOP) task, where the model must distinguish the correct order of two consecutive text segments from their reversed order. ELECTRA (2020) applied the idea of generative adversarial networks to the MLM task. Instead of masking out tokens, a small language model generates random plausible substitutions, and a larger network identify these replaced tokens. The small model aims to fool the large model. DeBERTa (2020) is a significant architectural variant, with disentangled attention. Its key idea is to treat the positional and token encodings separately throughout the attention mechanism. Instead of combining the positional encoding (x_{position}) and token encoding (x_{token}) into a single input vector ($x_{\text{input}} = x_{\text{position}} + x_{\text{token}}$), DeBERTa keeps them separate as a tuple: $(x_{\text{position}}, x_{\text{token}})$. Then, at each self-attention layer, DeBERTa computes three distinct attention matrices, rather than the single attention matrix used in BERT: